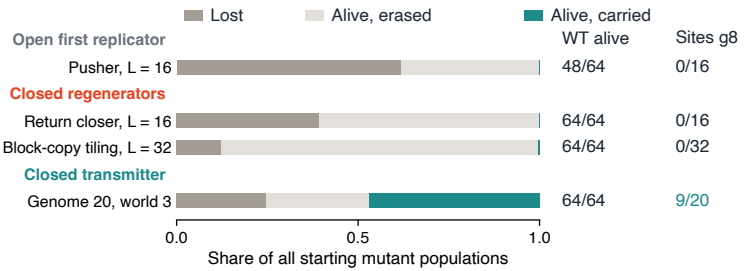
a **Branching populations, generation eight**

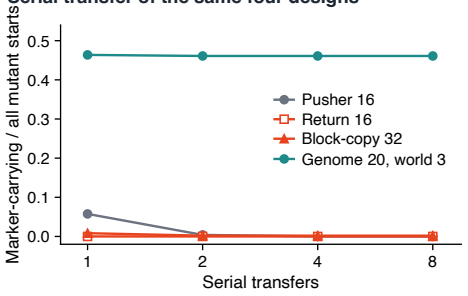


Bars: all mutant starts. Carrying means ≥ half the population retains its marker.

A site needs ≥ half its tested values to qualify; rare carriers can score zero sites.

WT: 64 unmutated starts. Original and corrected Z80 agree; pusher retention prediction contradicted.

b **Serial transfer of the same four designs**

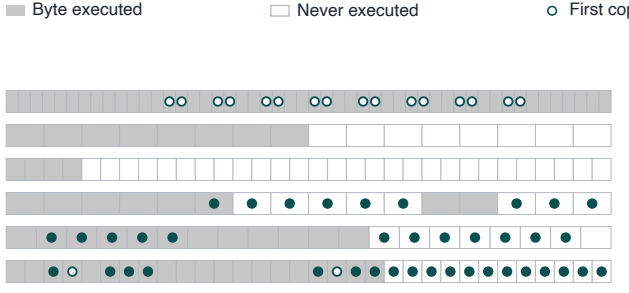


Marker = aligned 5-byte mutant window; all starts pooled.

255 values/site, 16 lines/value; original Z80 only.

Three curves coincide near zero at transfer 8.

c **Serial sites: one box per byte; each track scaled to its tape**

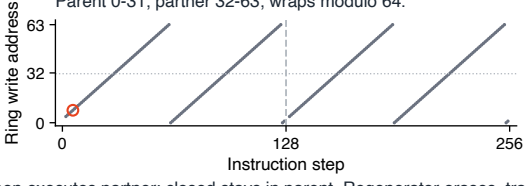


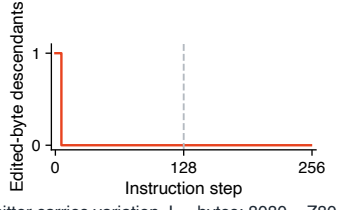
	Fetched bytes	Sites at transfer 1	Sites at transfer 8	Transfer-8 not fetched
Pusher, L = 50	50/50	16	0	0
Return closer, L = 16	8/16	0	0	0
Block-copy tiling, L = 32	4/32	0	0	0
Lethal-tar closer, L = 16	8/16	9	9	8
Genome 20, world 4	12/20	12	12	7
Constructed 8080, L = 32	20/32	21	19	12

Grey: opcode fetched in any of 64 random-partner executions of the unmutated tape (128 instructions each).
Serial site: ≥ half surviving lines carry above background, for ≥ half identifiable alternative values.

d **Block-copy regenerator from a-c: one erasure lineage**

Parent 0-31; partner 32-63; wraps modulo 64.





Initial label: edited byte 8 only (0xA7).

Erasing write at instruction 6:

source 0 (0x04) to destination 8 (0xA7).

Circled write removes last labelled byte.

128: offspring becomes parent; CPU resets.

Ancestry of one byte; b uses a 5-byte window.

All-zero partner; clock spans both encounters.

Open executes partner; closed stays in parent. Regenerator erases, transmitter carries variation. L = bytes; 8080 = Z80 subset.